

Linear Regression

Technical Notes

Rodrigo Kang

Table of contents

1	Machine Learning Modelling	1
1.1	Learning Paradigms	2
1.2	Combining Modelling Approaches	2
2	Notation	2
2.1	Predictors	3
2.2	Response and Prediction	3
3	Supervised Learning	5
3.1	Dataset	5
3.2	Underlying Relationship	6
3.3	Estimating the Relationship	6
3.4	Model Fitting	7

1 Machine Learning Modelling

A central part of data science is finding patterns in data. To do that, we try to identify the underlying structure or relationships in the data and represent them through models. This process is called modelling. A model is an abstract and simplified representation of some aspect of reality.

Broadly speaking, there are two ways we can approach this: from first principles or from data:

$$\text{Modelling} \longrightarrow \begin{cases} \text{First-Principles Modelling — Deductive} \\ \text{Data-Driven Modelling — Inductive} \end{cases} \quad (1)$$

Machine learning belongs broadly to the data-driven approach.

1.1 Learning Paradigms

Depending on the information available and the learning problem, we can distinguish between supervised, unsupervised and reinforcement learning.

In supervised learning, we have labelled data, meaning that each observation includes a response or target variable. If the response is quantitative, we typically have a regression problem; if it is categorical, we have a classification problem.

In unsupervised learning, there is no response variable. Instead, we're interested in discovering structure in the data itself, for example through clustering or dimensionality reduction.

Reinforcement learning is different from both. An agent learns by interacting with an environment, taking actions and receiving rewards, with the goal of learning a policy that maximises cumulative reward over time.

$$\text{ML} \longrightarrow \begin{cases} \text{SL} & \begin{cases} \text{Regression} & Y \text{ quantitative} \\ \text{Classification} & Y \text{ categorical} \end{cases} \\ \text{UL} & \text{no response } Y \\ \text{RL} & (s_t, a_t, r_t) \end{cases} \quad (2)$$

1.2 Combining Modelling Approaches

In practice, these approaches don't have to be mutually exclusive. In one of the projects I've worked on, for example, we were interested in predicting microbiologically influenced corrosion, or MIC. We combined mechanistic models, such as Lotka–Volterra and Monod models, with data-driven regression and classification approaches.

The same general idea appears in areas such as weather forecasting, where physical models of atmospheric dynamics can be combined with machine-learning methods.

2 Notation

Throughout these notes, N denotes the number of observations and M the number of predictors.

2.1 Predictors

The predictor variables are denoted by

$$X = (X_1, X_2, \dots, X_M). \quad (3)$$

For the i -th observation, the predictor values are collected in the vector

$$\mathbf{x}_i = \begin{pmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{iM} \end{pmatrix} = (x_{i1}, x_{i2}, \dots, x_{iM})^T \in \mathbb{R}^M, \quad (4)$$

where x_{ij} denotes the observed value of predictor j for observation i .

The complete set of predictors can then be arranged into the $N \times M$ matrix

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_N^T \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1M} \\ x_{21} & x_{22} & \cdots & x_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N1} & x_{N2} & \cdots & x_{NM} \end{pmatrix} \in \mathbb{R}^{N \times M}. \quad (5)$$

Each row of $\mathbf{X}$ represents one observation, while each column represents one predictor. In particular, $\mathbf{x}_i^T$ is the i -th row of the predictor matrix.

2.2 Response and Prediction

For a quantitative response, the response variable is denoted by

$$Y \in \mathbb{R}. \quad (6)$$

The observed response for observation i is denoted by y_i , and the complete set of responses is collected in the vector

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix} \in \mathbb{R}^N. \quad (7)$$

For a categorical response, the response variable is denoted by

$$C \in \mathcal{C}, \quad (8)$$

where $\mathcal{C}$ is the set of possible classes. For example, for a problem with K classes,

$$\mathcal{C} = \{1, 2, \dots, K\}. \quad (9)$$

The observed class for observation i is denoted by $c_i \in \mathcal{C}$, and the complete set of observed class labels is collected in

$$\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} \in \mathcal{C}^N. \quad (10)$$

Given a new predictor vector

$$\mathbf{x} \in \mathbb{R}^M, \quad (11)$$

a regression model produces an estimate of the quantitative response,

$$\hat{Y} = \hat{\varphi}(\mathbf{x}), \quad (12)$$

where

$$\hat{\varphi} : \mathbb{R}^M \rightarrow \mathbb{R}. \quad (13)$$

Similarly, a classification model produces an estimate of the categorical response,

$$\hat{C} = \hat{\varphi}(\mathbf{x}), \quad (14)$$

where

$$\hat{\varphi} : \mathbb{R}^M \rightarrow \mathcal{C}. \quad (15)$$

3 Supervised Learning

3.1 Dataset

In supervised learning, each observation is paired with its corresponding response. For a regression problem, the dataset is denoted by

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N, \quad (16)$$

where $\mathbf{x}_i \in \mathbb{R}^M$ is the predictor vector for observation i and $y_i \in \mathbb{R}$ is its corresponding quantitative response.

For a classification problem, the dataset is denoted by

$$\mathcal{D} = \{(\mathbf{x}_i, c_i)\}_{i=1}^N, \quad (17)$$

where $c_i \in \mathcal{C}$ is the class label associated with observation i .

When the dataset is partitioned for model development and evaluation, the corresponding subsets are denoted by

$$\mathcal{D}_{\text{train}}, \quad \mathcal{D}_{\text{val}}, \quad \mathcal{D}_{\text{test}}, \quad (18)$$

for the training, validation, and test sets, respectively.

The structure of a regression dataset is illustrated in Table 1.

Observation	X_1	X_2	$\cdots$	X_M	Y
1	x_{11}	x_{12}	$\cdots$	x_{1M}	y_1
2	x_{21}	x_{22}	$\cdots$	x_{2M}	y_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
N	x_{N1}	x_{N2}	$\cdots$	x_{NM}	y_N

Table 1: Tabular representation of a supervised regression dataset.

The columns $X_1, X_2, \dots, X_M$ correspond to the predictors, while Y corresponds to the response variable. Each row represents one observation, so that row i corresponds to the pair

$$(\mathbf{x}_i, y_i), \quad (19)$$

where $\mathbf{x}_i$ is defined in Equation 4.

For a classification problem, the same structure applies, replacing the quantitative response Y and its observed values y_i with the categorical response C and class labels c_i , respectively.

3.2 Underlying Relationship

For a regression problem, we can represent the relationship between the response and the predictors as

$$Y = \varphi(X) + \varepsilon, \quad (20)$$

where φ represents the systematic relationship between the predictors and the response, and ε represents the random component that cannot be explained by the predictors.

For the i -th observation,

$$y_i = \varphi(\mathbf{x}_i) + \varepsilon_i. \quad (21)$$

The goal is to use the available data to estimate the unknown relationship φ . The resulting estimate is denoted by $\hat{\varphi}$.

Given a new predictor vector $\mathbf{x}$, the corresponding prediction is then

$$\hat{Y} = \hat{\varphi}(\mathbf{x}). \quad (22)$$

For classification, the same general goal applies: we want to learn the relationship between the predictors and the categorical response, although the relationship is not generally expressed through the additive form in Equation 20.

3.3 Estimating the Relationship

There are different ways of approaching the estimation of φ . At a high level, we can distinguish between prespecifying a functional form and allowing a more flexible relationship to be learned from the data:

$$\text{Estimating } \varphi \longrightarrow \begin{cases} \text{Prespecifying a Functional Form} \\ \text{Learning a Flexible Relationship from Data.} \end{cases} \quad (23)$$

In the first approach, the general structure of the relationship is specified before fitting the model. For example, we may assume a linear relationship,

$$\varphi_{\theta}(\mathbf{x}) = \theta_0 + \sum_{j=1}^M \theta_j x_j. \quad (24)$$

The learning problem then reduces to estimating the parameters

$$\theta = (\theta_0, \theta_1, \dots, \theta_M)^T. \quad (25)$$

Linear regression is a canonical example of this approach. Polynomial regression and other explicitly parameterised functional forms follow the same general idea.

Alternatively, we may use a more flexible model class and allow the data to determine a larger part of the structure of the relationship. Examples include k-nearest neighbours, decision trees, random forests, boosting methods, and neural networks.

These two approaches should not be interpreted as assumptions versus no assumptions. Every learning method imposes some restrictions or preferences on the relationships that can be learned. This is known as **inductive bias**.

For example, linear regression restricts the model to linear relationships, k-nearest neighbours relies on the idea that nearby observations in predictor space should tend to have similar responses, decision trees represent relationships through recursive partitions of the predictor space, and neural networks restrict the learned relationship through their architecture and composition of transformations.

3.4 Model Fitting

Once a model has been specified, we need a criterion for determining how well it fits the observed data. This is usually expressed through a cost function, also commonly referred to as a loss function or objective function, which we denote by J .

The general strategy is to find the model that minimises this function:

$$\hat{\varphi} = \arg \min_{\varphi} J[Y, \varphi(X)]. \quad (26)$$

When the model is parameterised by a vector of parameters θ , this can instead be written as

$$\hat{\theta} = \arg \min_{\theta} J(\theta). \quad (27)$$

The particular form of J depends on the problem and determines what constitutes an optimal prediction. Consider first the L_2 loss, where prediction errors are measured by their squared magnitude. The expected cost associated with a function φ is

$$J(\varphi) = \mathbb{E} \left[(Y - \varphi(X))^2 \right]. \quad (28)$$

Using the definition of expectation with respect to the joint distribution of X and Y , this can also be written as

$$J(\varphi) = \int [y - \varphi(x)]^2 P(dx, dy), \quad (29)$$

where $P(dx, dy)$ denotes integration with respect to the joint probability distribution of (X, Y) . If X and Y have a joint probability density $p_{X,Y}(x, y)$, the same expression becomes

$$J(\varphi) = \int_{\mathbb{R}^M} \int_{\mathbb{R}} [y - \varphi(x)]^2 p_{X,Y}(x, y) dy dx. \quad (30)$$

By conditioning on X and using the law of iterated expectation, Equation 28 can be written as

$$J(\varphi) = \mathbb{E}_X \left[\mathbb{E}_{Y|X} \left[(Y - \varphi(X))^2 \mid X \right] \right]. \quad (31)$$

The expected squared error can therefore be minimised pointwise. For a fixed value $X = x$, the quantity $\varphi(x)$ is simply a scalar. Let $\alpha \in \mathbb{R}$ represent a candidate value for this prediction. We then seek

$$\varphi(x) = \arg \min_{\alpha \in \mathbb{R}} \mathbb{E} [(Y - \alpha)^2 \mid X = x]. \quad (32)$$

To solve this minimisation explicitly, define

$$J(\alpha) = \mathbb{E} [(Y - \alpha)^2 \mid X = x]. \quad (33)$$

Expanding the squared term and using the linearity of expectation gives

$$J(\alpha) = \mathbb{E} [Y^2 \mid X = x] - 2\alpha \mathbb{E} [Y \mid X = x] + \alpha^2. \quad (34)$$

Differentiating with respect to α ,

$$\frac{dJ}{d\alpha} = -2\mathbb{E}[Y \mid X = x] + 2\alpha. \quad (35)$$

Setting the derivative equal to zero gives

$$\alpha = \mathbb{E}[Y \mid X = x]. \quad (36)$$

Since

$$\frac{d^2 J}{d\alpha^2} = 2 > 0,$$

this stationary point is a minimum. Therefore,

$$\varphi(x) = \mathbb{E}[Y \mid X = x]. \quad (37)$$

Under L_2 loss, the optimal prediction is therefore the **conditional mean** of Y given $X=x$. This conditional expectation is commonly referred to as the **regression function**. Importantly, this is a general result: no linear relationship between X and Y has been assumed.

The choice of loss function changes what constitutes an optimal prediction. To see this, consider instead the L_1 loss, where prediction errors are measured by their absolute magnitude:

$$J(\varphi) = \mathbb{E}[|Y - \varphi(X)|]. \quad (38)$$

As before, conditioning on X allows the problem to be minimised pointwise:

$$\varphi(x) = \arg \min_{\alpha \in \mathbb{R}} \mathbb{E}[|Y - \alpha| \mid X = x]. \quad (39)$$

For a fixed $X = x$, define

$$J(\alpha) = \mathbb{E}[|Y - \alpha| \mid X = x]. \quad (40)$$

Assuming that the conditional distribution of Y given $X = x$ has a density $p(y \mid x)$, the expectation can be split at α :

$$J(\alpha) = \int_{-\infty}^{\alpha} (\alpha - y)p(y \mid x) dy + \int_{\alpha}^{\infty} (y - \alpha)p(y \mid x) dy. \quad (41)$$

Differentiating with respect to α gives

$$\frac{dJ}{d\alpha} = \int_{-\infty}^{\alpha} p(y | x) dy - \int_{\alpha}^{\infty} p(y | x) dy. \quad (42)$$

Introducing the conditional cumulative distribution function

$$F_{Y|X}(\alpha | x) = P(Y \leq \alpha | X = x), \quad (43)$$

we obtain

$$\frac{dJ}{d\alpha} = 2F_{Y|X}(\alpha | x) - 1. \quad (44)$$

At the minimum,

$$F_{Y|X}(\alpha | x) = \frac{1}{2}, \quad (45)$$

which means that the optimal value of α is the median of the conditional distribution. Therefore,

$$\varphi(x) = \text{median}(Y | X = x). \quad (46)$$

Under L_1 loss, the optimal prediction is therefore the **conditional median** rather than the conditional mean.

The two results can be summarised as

$$\begin{aligned} L_2 &\longrightarrow \mathbb{E}[Y | X = x], \\ L_1 &\longrightarrow \text{median}(Y | X = x). \end{aligned} \quad (47)$$

Thus, the choice of cost function does more than determine how prediction errors are penalised: it determines which property of the conditional distribution is optimal to predict.